Mass–Spring–Damper State-Space Model

Governing Equation

The mass–spring–damper system is governed by:

$$m\ddot{x} + b\dot{x} + kx = F$$

where:

- m = mass
- b = damping coefficient
- k = spring stiffness
- x = displacement
- F = applied force

Rearranging:

$$\ddot{x} = -\frac{k}{m}x - \frac{b}{m}\dot{x} + \frac{1}{m}F$$

Definition of States

Two states are selected:

$$x_1 = x$$

$$x_2 = \dot{x}$$

Therefore:

$$\dot{x}_1 = x_2$$

and:

$$\dot{x}_2 = -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{1}{m}F$$

Thus:

For the selected values ($m = b = k = 1$), we obtain:

$$\boldsymbol{A} = \begin{bmatrix} \mathbf{0} & \mathbf{1} \\ \mathbf{-1} & \mathbf{-1} \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} \mathbf{0} \\ \mathbf{1} \end{bmatrix}, \quad \boldsymbol{C} = [\mathbf{1} \quad \mathbf{0}], \quad \boldsymbol{D} = [\mathbf{0}]$$

Transfer-Function Representation

The mass–spring–damper transfer function is:

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}$$

For $m = b = k = 1$, the transfer function becomes:

$$G(s) = \frac{1}{s^2 + s + 1}$$

Transition to Aircraft Pitch Control

The aircraft plant transfer function is:

$$G(s) = \frac{-1.282s + 1.282}{s^3 + 1.935s^2 + 0.987s + 0.179}$$

Aircraft Pitch State-Space Model

The state-space representation obtained in Controllable Canonical Form is:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -0.179 & -0.987 & -1.935 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$C = [1.282 \quad -1.282 \quad 0], \quad D = [0]$$

Therefore, the state-equation dynamic model is:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -0.179 & -0.987 & -1.935 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \delta_e$$

and the output equation for the pitch angle θ is $\theta = [1.282 \quad -1.282 \quad 0]x$